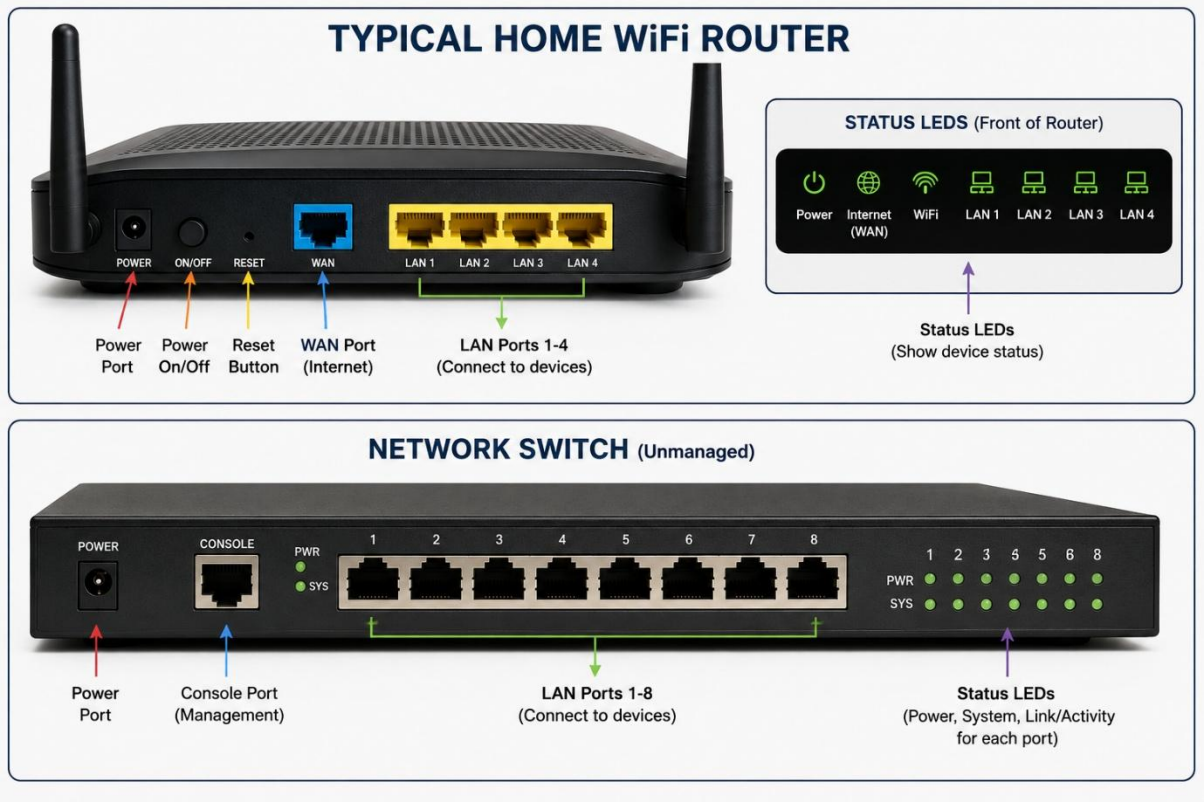
1.

Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



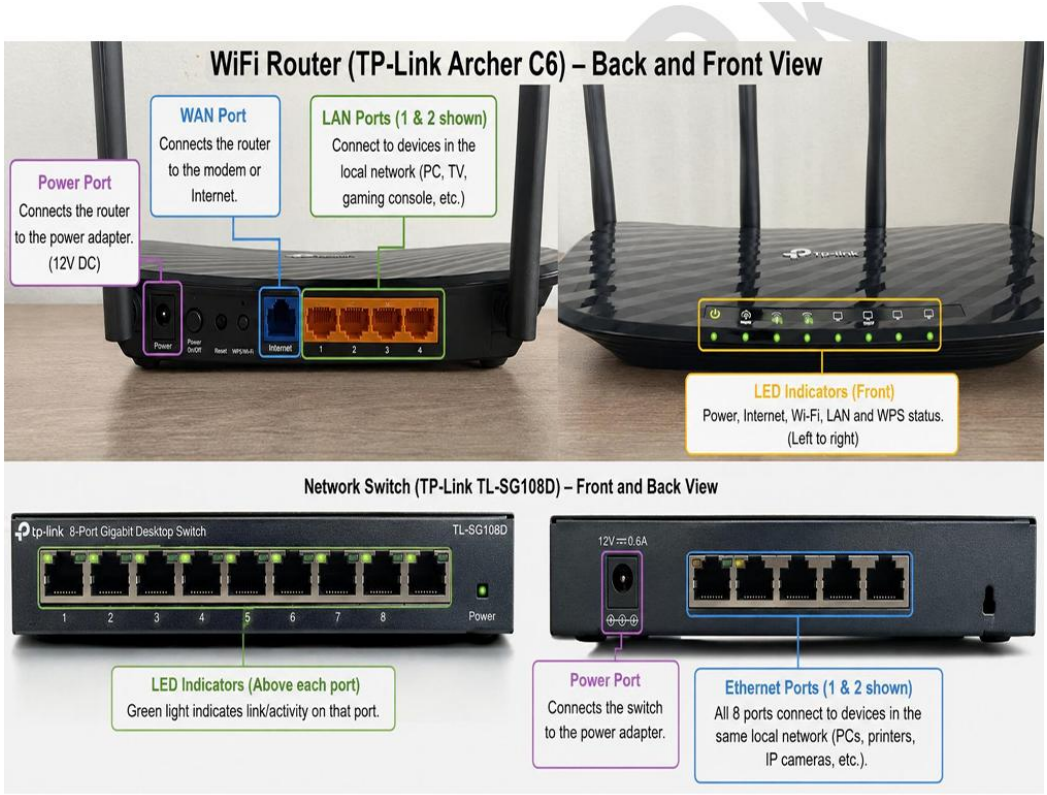
2.

List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Difference	Router	Switch
1. Main Function	Connects **different networks** and sends data between them.	Connects **devices within the same network**.
2. Address Used	Uses **IP addresses** to decide where data should go.	Uses **MAC addresses** to send data to the correct device.
3. Network Type	Connects a **LAN to another network**, such as the Internet (WAN).	Mainly connects devices in a **LAN**, such as computers and printers.

3.

Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



4.

Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Simple Scenario

Imagine a small gaming café with **8 gaming PCs** and an internet connection from an **ISP**.

- First, I would connect the **ISP line to the router**.
- The **router** provides internet access and manages the network.
- Then, I would connect the router to a **network switch** using an Ethernet cable.
- Finally, I would connect all **8 gaming PCs to the switch** using Ethernet cables.

Connection:

ISP → Router → Switch → 8 Gaming PCs

Justification:
The **router must be connected first** because it connects the café network to the internet. The **switch** is then used to connect multiple PCs together. Using Ethernet cables gives the gaming PCs a more **stable and faster connection**, which is useful for multiplayer gaming and IPL streaming.